Matematikk 1000 Esmir Ahat Studentnummer: 405121

Matematikk 1000 Oblig 2

$f(x) = e^{-x/4} \cdot \tan^{-1} x$ | maksimal punkt

$u = e^{-x/4} \quad v = \tan^{-1}(x)$

$u' = -\frac{1}{4} e^{-x/4} \quad v' = \frac{1}{1+x^2}$

$f'(x) = v'u + vu'$

$= \frac{1}{1+x^2} \cdot e^{-x/4} + \tan^{-1}(x) \cdot \left(-\frac{1}{4} e^{-x/4}\right)$

$= e^{-x/4} \left(\frac{1}{1+x^2} - \frac{1}{4} \tan^{-1}(x)\right)$

$f'(x) = 0$

$0 = e^{-\frac{x}{4}} \left(\frac{1}{1+x^2} - \frac{1}{4} \tan^{-1}(x)\right)$

$e^{-\frac{x}{4}} > 0$ for alle x-verdier tar vi det ikke fra uttrykket

dvs $0 = \frac{1}{1+x^2} \cdot 4 - \frac{1}{4} \tan^{-1}(x) \cdot 4$

$0 = \frac{4}{1+x} - \tan^{-1} x$

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```
import numpy as np
import matplotlib.pyplot as plt
import sympy as sp

x = sp.symbols('x')
f = sp.atan(x) - 4 / (x**2 + 1)
XforT = sp.nsolve(f, 1)

String1 = "X verdi for Toppunktet er"
print(String1, round(XforT,4),)
#X verdi for Toppunktet er 1.6907

Xverdi = 1.6907
y = np.exp(-Xverdi/4)*np.arctan(Xverdi)

String2 = "Y verdi for Toppunktet er"
print(String2, round(y,4))
print("(x,y) = ", "(", round(XforT,4), ",",round(y,4),")")
# Y verdi for Toppunktet er 0.6793

#plt del
def f(x):
    return np.exp(-x/4)*np.arctan(x)

Xintervaller = np.linspace(-2,5,500)
Yintervaller = f(Xintervaller)

#Toppunkt
Xmaks = 1.6907
Ymaks = f(Xmaks)

plt.plot(Xintervaller, Yintervaller, label='f(x)')
plt.scatter(Xmaks, Ymaks, color='red', label='Toppunkt')
plt.legend()
plt.grid()
plt.xlabel('x')
plt.ylabel('f(x)')
plt.title('Graf av f(x) = e^(-x/4) * arctan(x)')
plt.show()
```